

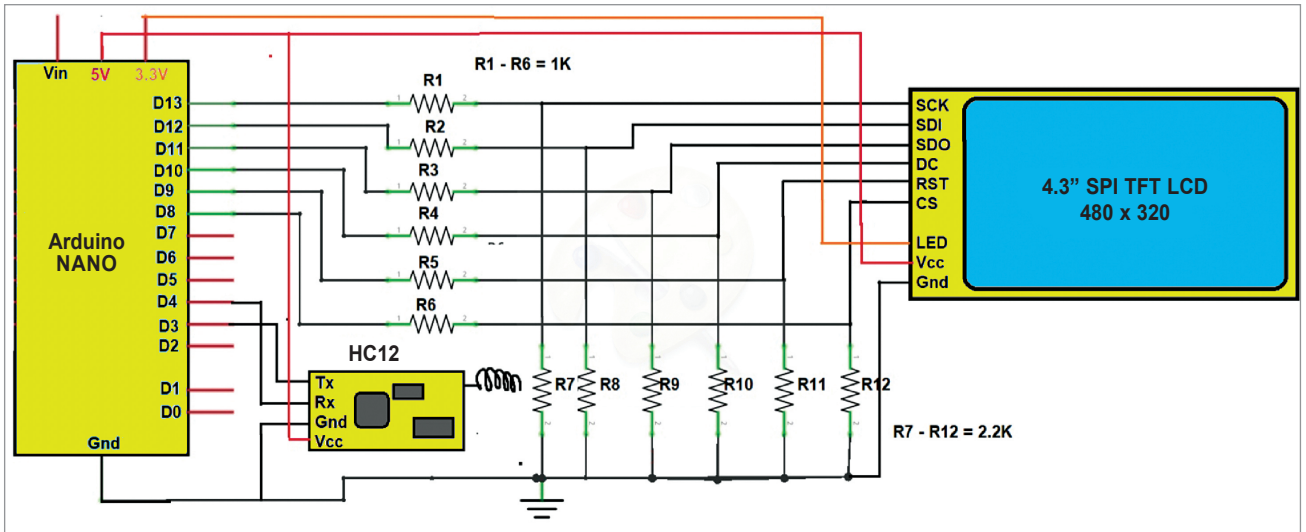


Fig. 5: Receiver circuit

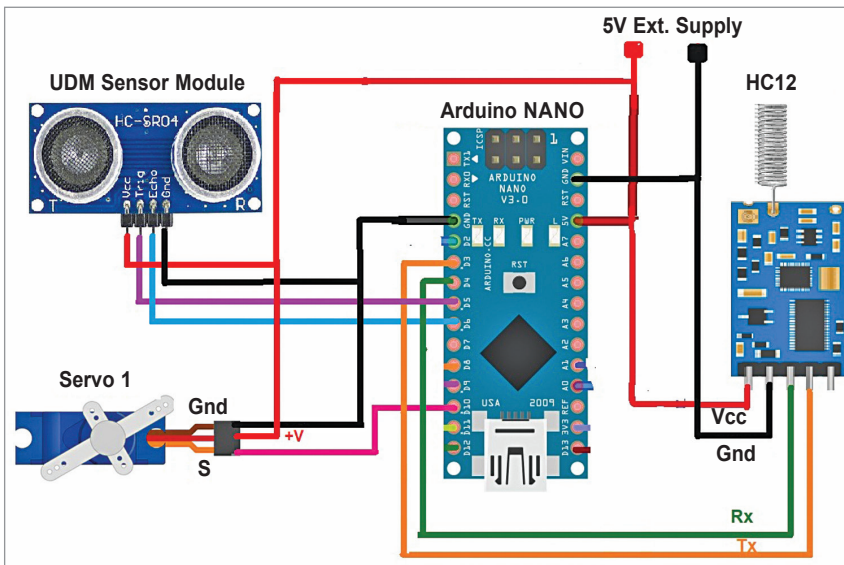


Fig. 6: Wiring diagram of transmitter

Arduino board.

The HC-12 module has four interfacing pins: Vcc, Gnd, Tx, and Rx. The Vcc pin is connected to the 5V output pin of the Arduino board, and the Gnd pin is connected to the common circuit ground. The Tx and Rx pins are connected to digital pins D3 and D4 of the Arduino board, respectively.

**Radar receiver section.** The receiver section consists of three components: the Arduino Nano, the RF

transceiver, and the 10.9cm (4.3-inch) TFT display. The block diagram of the receiver section is shown in Fig. 4. A brief description of the components used is given here.

The Arduino Nano is again used here. The RF transceiver is used to serially receive data as distance and angle of an object and sends it to the Arduino Nano. This is the main component, which receives data from the RF transceiver and displays the distance and angle of the object on the

TFT display. It also displays the radar screen dial and the animation of the object scanning process. The TFT display acts as the radar monitor, showing a radar dial (semi-circle) display and continuously scanning process similar to a radar dial. It also displays the distance and angle of any detected object.

**Circuit diagram of radar receiver section.** Fig. 5 shows the circuit diagram of the receiver section. This circuit is built around the TFT LCD, the HC-12 module, and the Arduino Nano. Instead of the Arduino Nano, an Arduino Uno can be used, as shown in the prototype (Fig. 8b). This substitution does not affect the operation of the device.

As seen in the transmitter figure, the HC-SR04 module is mounted on the servo motor shaft. There is a 0-180° angle display dial attached to the servo motor, and an angle pointer is attached to the motor shaft. As the servo motor rotates, its angle of rotation is indicated on the dial. Since the HC-SR04 sensor is also attached to the same shaft, this angle indicates the angle of any detected object.

## Operation of the circuit

Initially, both the radar transmitter and receiver are off. First, the receiver